

LAB 5: CI/CD Pipeline using Jenkins, Docker & Kubernetes

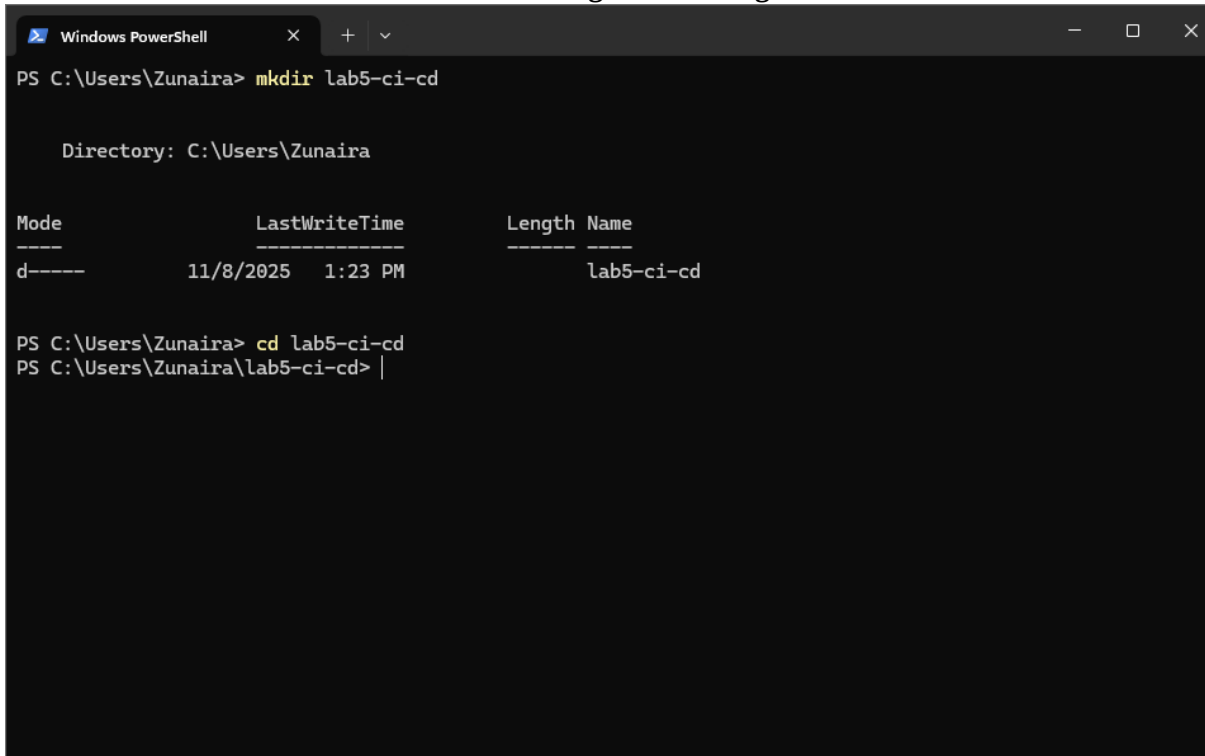
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BSSE23058

Objectives:

The main goal is to use Docker Compose to run an application with two containers. We will create a Python web app and a MySQL database that work together. We'll write a single docker-compose.yml file to connect and run both. This lab also teaches us how to use environment variables and save database data.

Screenshot #1: Show the terminal after creating and entering the folder.



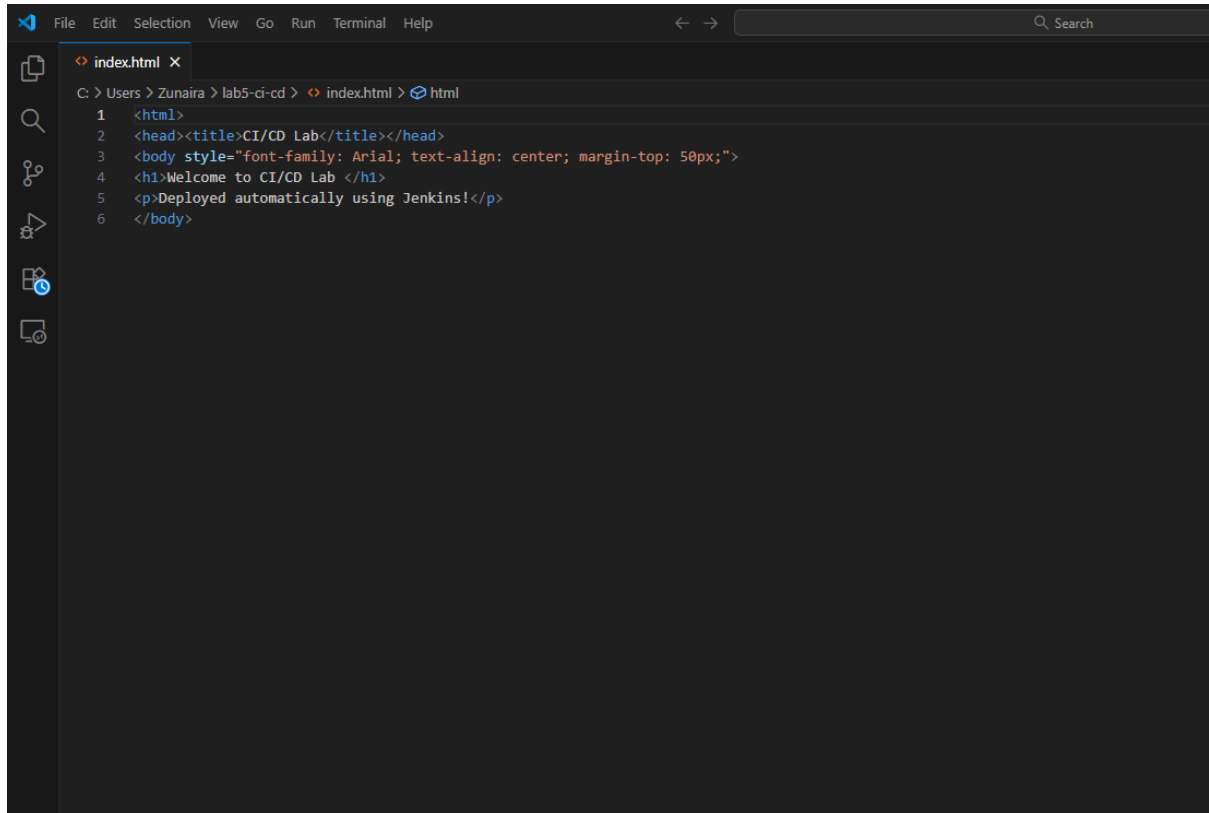
```
Windows PowerShell
PS C:\Users\Zunaira> mkdir lab5-ci-cd

Directory: C:\Users\Zunaira

Mode                LastWriteTime         Length Name
----                -
d-----           11/8/2025   1:23 PM             lab5-ci-cd

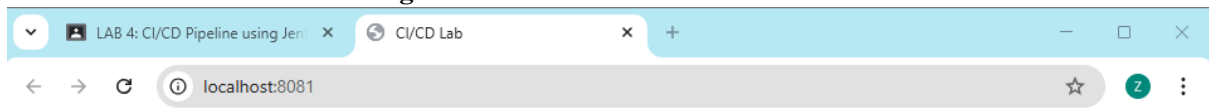
PS C:\Users\Zunaira> cd lab5-ci-cd
PS C:\Users\Zunaira\lab5-ci-cd> |
```

Screenshot #2: Show this code in VS Code.

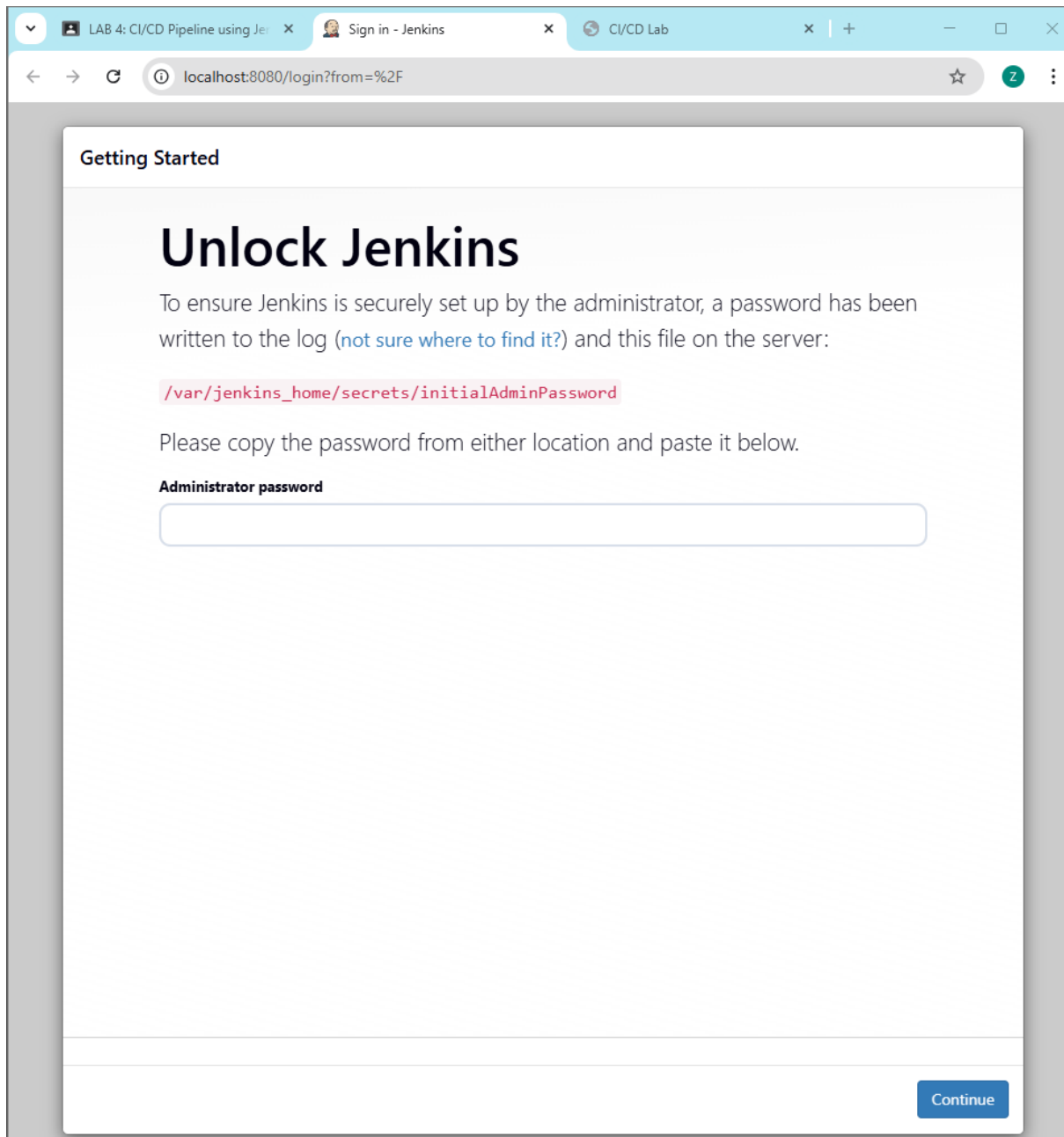
A screenshot of the Visual Studio Code editor interface. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. The left sidebar shows icons for Explorer, Search, Source Control, Run and Debug, and Remote Explorer. The main editor area displays a file named 'index.html' with the following HTML code:

```
1 <html>
2 <head><title>CI/CD Lab</title></head>
3 <body style="font-family: Arial; text-align: center; margin-top: 50px;">
4 <h1>Welcome to CI/CD Lab </h1>
5 <p>Deployed automatically using Jenkins!</p>
6 </body>
```

Screenshot #3: Browser showing “Welcome to CI/CD Lab



Screenshot #4: Jenkins setup screen with password prompt.



Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/jenkins_home/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

Continue

```
Windows PowerShell
PS C:\Users\Zunaira\lab5-ci-cd> docker run -d -p 8080:8080 -p 50000:50000 --name jenkins -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
cae3b572364a: Pull complete
cc05fa07d253: Pull complete
e29665228ac2: Pull complete
7c2b9fc47dae: Pull complete
eba243d676e4: Pull complete
606dcc9d6add: Pull complete
11c82e82e8c5: Pull complete
04e220b291b8: Pull complete
9e58f885f660: Pull complete
51148860bddf: Pull complete
6d8ebcba18e6: Pull complete
b9bcce170b58: Pull complete
Digest: sha256:f2519b99350faeaaef30e3b8695cd6261a5d571c859ec37c7ce47e5a241458d
Status: Downloaded newer image for jenkins/jenkins:lts
26d20e855f4759cc28b80cb5e9876154f900fb2279a05b44adef543ad664cbb9
PS C:\Users\Zunaira\lab5-ci-cd> docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword

b513d4b9bd0f400790c98e2abfda489e
PS C:\Users\Zunaira\lab5-ci-cd> ^C
PS C:\Users\Zunaira\lab5-ci-cd>
```

Screenshot #5: Show installed plugins.

Jenkins

Manage Jenkins

Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Q Search installed plugins

1

Adds a dark theme to Jenkins, respecting browser and OS themes if configured.
Report an issue with this plugin

99

☒

☐

Display URL API 2.217.ua.6b_d684cc74b_

97

☐

☐

Docker Commons Plugin 457.v0f62a_94f11a_3

97

☐

☐

Docker Pipeline 124.103c75d42b_25a2_7

94

☒

☐

Build and use Docker containers from pipelines.
Report an issue with this plugin

100

☐

☐

Durable Task Plugin 635.v373cef34b_5e

100

☐

☐

Library offering an extension point for processes which can run outside of Jenkins yet be monitored.
Report an issue with this plugin

96

☐

☐

ECharts API Plugin 6.0.0-1165.wd1283a_3e3764

100

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☐

Provides ECharts for Jenkins Plugins, a JavaScript visualization tool to create intuitive, interactive, and highly-customizable charts.
Report an issue with this plugin

100

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☐

EDDSA API Plugin 0.3.0-1-19vc432d923e5ee

100

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EDDSA api plugin for use by other Jenkins plugins.
Note:Modern (> 15) JDK variants natively support EdDSA signatures via JCE but some libraries still target JDK 11 and have a need of this library.
Report an issue with this plugin

100

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Email Extension Plugin 1933.v45cec755423f

100

☒

☐

This plugin is a replacement for Jenkins's email publisher. It allows to configure every aspect of email notifications: when an email is sent, who should receive it and what the email says.
Report an issue with this plugin

100

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☐

Folders Plugin 6.1073.ava_7888eb_dcf514

100

☐

☐

This plugin allows users to create "folders" to organize jobs. Users can define custom taxonomies (like by project type, organization type etc). Folders are nestable and you can define views within folders. Maintained by CloudBees, Inc.
Report an issue with this plugin

Jenkins

Manage Jenkins

Plugins

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Download progress

Q kuber

1

Name	Health	Enabled	
Kubernetes CLI Plugin 1364.vadebf8cb8b823	100	☒	☐
Configure kubectl for Kubernetes Report an issue with this plugin			
Kubernetes Client API Plugin 7.3.1-256.v788a_0b_787114	96	☐	☐
Kubernetes Client API plugin for use by other Jenkins plugins. Report an issue with this plugin			
Kubernetes Credentials Plugin 205.vd631a_b_0f71a_c	100	☐	☐
Common classes for Kubernetes credentials Report an issue with this plugin			

LAB 4: CI/CD Pipeline using Jen...

Installed plugins - Plugins - Jen...

CI/CD Lab

localhost:8080/manage/pluginManager/installed

Jenkins

Manage Jenkins

Plugins

Search

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Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Q. github

Name	Health	Enabled
GitHub API Plugin 1.330-492.v0941e_032db_2e_... This plugin provides GitHub API for other plugins. Report an issue with this plugin	100	☒
GitHub Branch Source Plugin 1917.v9ee8a_39b_3d0d Multibranch projects and organization folders from GitHub. Maintained by CloudBees, Inc. Report an issue with this plugin	100	☒
GitHub Integration Plugin v0.2 GitHub Integration Plugin for Jenkins	97	☒
GitHub plugin 1.45.0 This plugin integrates GitHub to Jenkins. Report an issue with this plugin	100	☒
Pipeline: GitHub Groovy Libraries 65.v20368be7727e Allows Pipeline Groovy libraries to be loaded on the fly from GitHub. Report an issue with this plugin	97	☒

REST API

Jenkins 2.528.1

24°C Sunny

Search

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1:48 PM

08-Nov-2023

Screenshot #6: Jenkins pipeline editor with code visible.

The screenshot shows the Jenkins Pipeline editor interface. The left sidebar contains navigation options: General, Triggers, Pipeline (selected), and Advanced. The main area displays a pipeline script in a text editor. The script defines a pipeline with two stages: 'Build' and 'Test'. The 'Build' stage includes a step to build a Docker image. The 'Test' stage includes a step to run a container and echo a message. Below the script, there is a checkbox for 'Use Groovy Sandbox' and a 'Pipeline Syntax' link. At the bottom, there are 'Save' and 'Apply' buttons.

```
1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         echo 'Building Docker image...'
7         sh 'docker build -t cicd-demo:v2 .'
8       }
9     }
10    stage('Test') {
11      steps {
12        echo 'Running container test...'
13        // This is a simple test. In a real scenario, this would
14        // be more complex, e.g., running unit or integration tests.
15        sh 'docker run --rm cicd-demo:v2 echo "Container is working fine"'
16      }
17    }
18  }
19 }
```

Screenshot #7: Jenkins pipeline success screen.

The screenshot shows the Jenkins pipeline success screen for the 'CI-CD-Lab5' pipeline. The left sidebar contains navigation options: Status (selected), Changes, Build Now, Configure, Delete Pipeline, Full Stage View, Stages, Rename, Pipeline Syntax, and Credentials. The main area displays the 'Stage View' with a table showing the stages and their durations. Below the table, there is a 'Permalinks' section with links to the last build, last stable build, last successful build, and last completed build. On the left, there is a 'Builds' section showing a list of builds with a filter and a dropdown menu.

Setup: Install Docker	Create Project Files	Build	Test	Deploy to Kubernetes
9min 19s	204ms	4s	3s	421ms

Average stage times: (full run time: ~9min 28s)

Nov-08 14:18 No Changes

9min 19s 204ms 4s 3s 421ms

Permalinks

- Last build (#10), 10 min ago
- Last stable build (#10), 10 min ago
- Last successful build (#10), 10 min ago
- Last completed build (#10), 10 min ago

Builds

Filter

Today

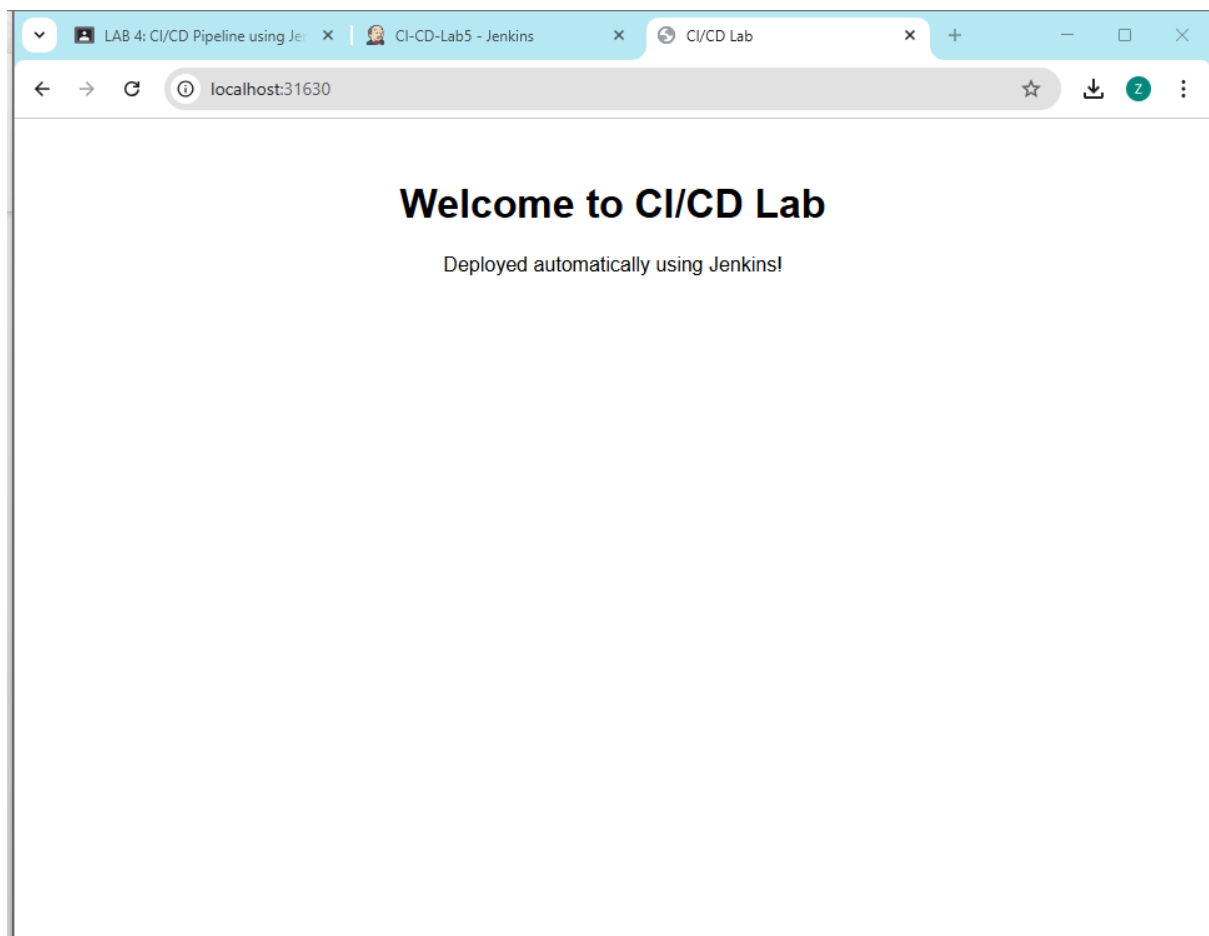
#10 9:16 AM

The screenshot shows a terminal window with the following commands and output:

```
PS C:\Users\Zunaira\lab5-ci-cd> kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
cicd-demo-deployment 2/2     2            2          90s
PS C:\Users\Zunaira\lab5-ci-cd> kubectl get pods
NAME                                     READY   STATUS    RESTARTS   AGE
cicd-demo-deployment-6c44f8ff5b-5lm8c 1/1     Running   0          92s
cicd-demo-deployment-6c44f8ff5b-t27t7 1/1     Running   0          92s
PS C:\Users\Zunaira\lab5-ci-cd>
```

Screenshot #8: Browser showing deployed page.

```
Windows PowerShell
PS C:\Users\Zunaira\lab5-ci-cd> kubectl get svc
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
cicd-demo-deployment              NodePort    10.106.169.113 <none>       80:31630/TCP     80s
kubernetes                         ClusterIP   10.96.0.1     <none>       443/TCP          6d20h
PS C:\Users\Zunaira\lab5-ci-cd> |
```



Observations

1. Jenkins is the "brain" that runs all the steps. It automatically tells Docker to build the app and tells Kubernetes to deploy it.
2. "Pipeline as Code" is just a script. We wrote the whole process (build, test, deploy) in one script. When it failed, we just had to fix the script and run it again.
3. The console log is essential. The pipeline failed many times. The only way to fix it was to read the "Console Output" to find errors like docker: not found or kubectl: not found.

Conclusion

This lab showed me how to build a real CI/CD pipeline. I learned how Jenkins connects Docker (for building) and Kubernetes (for deploying) all in one automated process. The most important thing I learned was how to read error logs, fix the pipeline script, and try again. This automation makes shipping software much faster and safer than doing everything by hand.